Epistemic tiers and the double-bridge chain

Tier	Currency	Bridge / interpretation
3 · Computational	Relative precision × excess error; leaky accumulation and access gate	Approximate inference bridge: not validated in vivo
2 · Information-theoretic	Bits per ignition event; event definition must be fixed	κ in ATP / event-information bit: uncalibrated
1 · Thermodynamic	Matched event energy in ATP or joules	Compare with independent accounting; Landauer is an erasure floor only

APGI 1.1 / P4F1 / Specification schematic

Paper 4 Fig 1: ignition information and erased information have different denominators; upward bridges remain unspecified.